

Machine Learning - Vectorization Numerical Questions

1. Given $X = [2, 3, 4]$ and $W = [1, 2, 3]$, calculate the dot product $X \cdot W$.
2. Given $X = [2, 4, 6]$, $W = [3, 2, 1]$, and $b = 5$, calculate $y = XW + b$.
3. Given $X = [[1, 2], [3, 4], [5, 6]]$, $W = [2, 3]$, and $b = 1$, calculate $Y = XW + b$.
4. Given $X = [2, 5, 3]$, $W = [4, 2, 1]$, and $b = 3$, calculate the predicted value y .
5. Given $X = [[1, 2], [2, 3], [3, 4], [4, 5]]$, $W = [2, 3]$, and $b = 2$, calculate Y .
6. Given $X = [[2, 1, 3], [4, 2, 1], [1, 3, 2]]$, $W = [3, 2, 4]$, and $b = 5$, calculate $Y = XW + b$.
7. A dataset contains 100 observations and 4 features. What are the dimensions of X , W , and Y in $Y = XW + b$?
8. Given $X = [[2, 4], [3, 5]]$, $W = [3, 2]$, and $b = 4$, calculate Y .
9. Given $X = [5, 2, 3]$, $W = [2, 4, 1]$, and $b = -3$, calculate $y = XW + b$.
10. Given $X = [1, 2, 3]$, $W = [4, 5, 6]$, and $b = 2$, calculate the vectorized prediction.